

Siddharth Sharma

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Biology / Computational Neuroscience)*

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Profile

Mathematical biologist and computational scientist working at the interface of stochastic dynamics, nonequilibrium statistical mechanics, and geometric deep learning. I develop methods to infer biological network structure from imaging and multimodal data, and build mechanistic models of neural population dynamics and control.

My current research focuses on the renal neurovascular–glomerular complex: converting 3D fluorescence light-sheet microscopy into anatomically grounded graphs (glomeruli, vessels, nerves, ducts), quantifying multiscale organization (including mesoscale communities and heavy-tailed connectivity patterns), and linking structure to physiology and disease-relevant perturbations.

A parallel thread of work addresses neural population criticality: extracting and modeling crackling-noise/avalanche statistics in whole-brain calcium imaging, and developing dynamical models that incorporate asymmetry (probability currents) and plasticity to reproduce observed scaling behavior.

Doctoral thesis (UMD) developed (i) inference in asymmetric nonequilibrium networks via latent-variable augmentation (Poisson and Pólya–Gamma) with EM optimization, and (ii) optimal transitions between nonequilibrium steady states of unequal network dimension using Gromov–Wasserstein optimal transport. Prior thesis work includes nanoparticle-superlattice phase behavior (M.S.E., published) and PDE-based surface diffusion (B.Tech., Best Thesis Award).

Research Interests

Network inference in biological systems; geometric deep learning (GNNs) for connectomics and spatial biology; stochastic differential equations and point processes; criticality and avalanche dynamics; optimal transport on metric measure spaces; multimodal registration (3D imaging + spatial transcriptomics); nonequilibrium steady states and asymmetric networks.

Selected Contributions

- HuBMAP-aligned mapping of human kidney neurovascular–nephron structure from 3D light-sheet microscopy, including graph construction and mesoscale organization analysis.
- Development of imaging-to-graph pipelines: 3D segmentation, feature extraction, and downstream network analysis for spatial biological maps.
- Whole-brain zebrafish sleep/sleep-deprivation analysis: calcium imaging preprocessing (motion correction + deconvolution) and avalanche/criticality statistics.
- Methods for nonequilibrium inference and control in asymmetric networks: latent-variable likelihood/EM constructions and optimal-transport-based driving between steady states.

Education

- 2023 **Ph.D., Biophysics Program (Home Dept.: Mathematics)**, *University of Maryland, College Park*, College Park, MD
Thesis: *Inference and Control in Networks Far From Equilibrium*. Advisor: Prof. Doron Levy.
- 2011 **M.Sc.(Engg.)**, **Chemical Engineering**, *Indian Institute of Science*, Bengaluru, India
Thesis work published as: *Effect of charge asymmetry and charge screening on structure of superlattices formed by oppositely charged colloidal particles*. *J. Chem. Phys.* (2012), doi:10.1063/1.3700226.
- 2008 **B.Tech., Chemical Engineering**, *Guru Gobind Singh Indraprastha University*, Delhi, India
Best Undergraduate Thesis Award; thesis on PDE-based modeling of surface diffusion.

Academic and Professional Appointments

- Nov 2023 – **Postdoctoral Research Associate (Systems Biology)**, *Washington University School of Medicine*,
Present Division of Nephrology, Jain Lab, St. Louis, MO
- Lead computational efforts for HuBMAP-aligned mapping of the human renal neurovascular–nephron complex using 3D fluorescence light-sheet microscopy and computer vision.
 - Build and analyze graph representations (glomeruli, vessels, nerves, ducts) and quantify mesoscale structure (e.g., community structure, spatial heterogeneity, heavy-tailed degree patterns) to connect anatomy to function and disease context.
 - Develop stochastic and network-dynamical models of nerve-mediated control in kidney, targeting robustness and functional bandwidth under stress and pathology-relevant perturbations.
 - Building a multimodal pipeline based on transformers and graph neural networks for combining histology tile embeddings and spatial transcriptomic features to predict spatial interaction potentials (cell–cell communication / ligand–receptor activity) and integrate these with 3D tissue structure.
 - Acquire 3D light-sheet volumes of cleared kidney tissue and translate volumetric data into anatomical graphs of glomeruli, nerves, and vessels for multiscale network analysis.
- Nov 2022 – **Postdoctoral Research Associate (Computational Neuroscience)**, *Washington University School*
Nov 2023 *of Medicine*, Department of Developmental Biology, St. Louis, MO
- Analyzed neuronal avalanche statistics in whole-brain zebrafish calcium imaging during natural and induced sleep and sleep deprivation.
 - Built motion-correction and deconvolution workflows for population-scale event extraction (including CalmAn-based processing).
 - Developed mechanistic models for critical dynamics in plastic neuronal populations with asymmetric transition structure and probability currents.
- Jul 2021 – **Tutor, Gene Golub SIAM Summer School (Mathematics of Deep Learning)**, *AIMS & SIAM*
Aug 2021 ○ Taught kernel methods and mathematical foundations of deep learning; supported applied coding modules for modern neural architectures.
- Jan 2014 – **Lecturer / Graduate Teaching Assistant**, *University of Maryland, College Park*, Department of
Aug 2022 Mathematics, College Park, MD
- Instructor of record: Differential Equations (MATH 246); Applied Probability & Statistics (MATH 214); Intro Statistics (STAT 100); Computing with MATLAB (MATH 206).
 - Teaching assistant: Calculus I–III; Differential Equations; Calculus for Life Sciences I–II.
- Jan 2021 – **Graduate Teaching Assistant**, *University of Maryland, College Park*, College of Information Studies,
May 2022 College Park, MD
- INST 326 (Object-Oriented Programming); INST 351 (Statistical Decision Theory).
- Jan 2012 – **Financial Software Consultant (MNREGA)**, *Ministry of Rural Development, Government of India*
Apr 2013 ○ Maintained and extended software supporting biometric-linked payment workflows for rural employment programs (C#, Java; distributed operations through rural hubs).

Aug 2008 – **Graduate Research Fellow**, *Indian Institute of Science*, Department of Chemical Engineering,
 Dec 2011 Bengaluru, India
 ○ Simulated phase behavior of charged nanoparticle superlattices; produced phase diagrams and stability analyses using DLVO-based modeling.

Apr 2008 – **Business Analyst (Energy and Materials)**, *McKinsey & Company*
 Aug 2008

--- Fellowships and Awards

2024–2025 Center for Regenerative Medicine Postdoctoral Research Fellowship, Washington University School of Medicine

2019 Hockmeyer Fellowship, Graduate School, University of Maryland

2018 Hockmeyer Fellowship, Graduate School, University of Maryland

2017 Instructor, NSF-sponsored REU on Disease Modeling, University of Maryland

2016 Summer Graduate Fellowship Award, Biophysics Program, University of Maryland

2014 All S.T.A.R Fellowship, College of Computer & Natural Science, University of Maryland

2008–2010 Research Fellowship, Indian Institute of Science

2007 Best Undergraduate Thesis Award (PDE modeling of surface diffusion), GGSIP University
 Undergraduate Research Fellow, National Physical Laboratory (India), Materials Science

--- Publications and Preprints

Peer-reviewed

McLaughlin, L., Zhang, B., **Sharma, S.**, et al. *Three-dimensional multiscalar neurovascular nephron connectivity map of the human kidney across the lifespan*. *Nature Communications* **16**, 5161 (2025). doi:10.1038/s41467-025-60435-8.

Sharma, S. and Punnathanam, S. *Effect of charge asymmetry and charge screening on structure of superlattices formed by oppositely charged colloidal particles*. *The Journal of Chemical Physics* **136**, 134506 (2012). doi:10.1063/1.3700226.

Submitted / under review

McLaughlin, L., Curic, M., **Sharma, S.**, Salamon, R., Yamaguchi, M., Lopez, L. M., El-Achkar, T., Gomez, A. R., & Jain, S. *NetTracer3D: A tool for network analysis of 3D spatial maps*. Manuscript submitted to *Nature Methods* (under review).

Preprints / manuscripts in preparation

Sharma, S. and Levy, D. *A solution to the inverse problem for asymmetric networks in absence of time-ordered data*. Preprint (2026).

Sharma, S. and Levy, D. *Optimal driving between nonequilibrium steady states of unequal dimensions*. Preprint (2026).

Sharma, S. *Geometric Hebbian learning of network degree distributions through whole-brain recordings*. Manuscript in preparation (2026).

Thesis / monograph

Sharma, S. *Inference and Control in Networks Far From Equilibrium*. Ph.D. Thesis, University of Maryland, College Park (2023).

Technical Skills

Imaging & Computer Vision	Fluorescence light-sheet microscopy workflows; 3D/volumetric image processing; image segmentation; registration and co-registration; feature extraction; microscopy-to-graph pipelines
ML / AI	Geometric deep learning (GNNs); representation learning; CNN-based approaches for imaging; time-series and neural data models
Neural Data	Calcium imaging analysis; motion correction; deconvolution (CalmAn); avalanche/criticality analysis
Mathematics & Theory	Probability; stochastic differential equations; optimal transport; inverse problems; nonequilibrium statistical mechanics
Networks & Modeling	Random graphs; complex networks; community detection; mechanistic stochastic models; ODE/PDE/agent-based simulation
Programming	Python (NumPy, SciPy, PyTorch, TensorFlow); MATLAB; C++; R; Spark

Teaching (Selected)

Instructor of record (UMD Mathematics): Differential Equations, Applied Probability/Statistics, Intro Statistics, MATLAB computing.

Tutor (AIMS/SIAM): kernel methods and mathematical foundations of deep learning.

Additional Interests

Condensed matter and phase transitions; nonequilibrium thermodynamics; cosmology and entropy production; yoga and endurance-style fitness.